higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WORKSHOP PRACTICE
NQF LEVEL 2

8 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
- Proper use of tools and equipment
- Correct use of safety equipment and clothes (PPE)
- Appropriate storage of equipment
- Appropriate storage of materials
- How to handle hazardous materials
- How to report injuries and problems
- How to respond to injuries at work (Any 5 × 1) (5)

1.2
- Workers who take shortcuts when they believe they are in control
- Bad housekeeping, poor stacking and wet floors
- Workers who have safety training, but do not put their training to into practice or insufficient training
- or lack of training
- Needs of the factory environment are not met, e.g. excessive noise levels, poor lighting, etc.
- Lack of work standards, e.g. failing to use PPE,
- or not providing workers with PPE
- Horse play. (Any 3 × 1) (3)

1.3
- Chemical hazards: acid, solvents and paint, gases and fumes
- Biological hazards: bacteria, viruses, and moulds
- Ergonomic hazards: poor work-station design, repetitive movement, awkward movement and poor posture
- Physical hazards: electric currents, heat, light, noise, radiation (Any 4 × 1) (4)

1.4
- Merchant Shipping Act (Marine)
- Mine Health And Safety Act (2)

1.5 Integrated Five Star System Standard (2)

1.6 Scaffolding is a temporary structure on the outside of a building (1)

1.7
- Wear a safety harness that is in good working order
- Tie off the lanyard of the safety harness above the head level
- Cordon off the area below you as warning for the people walking on the ground level
- Do not raise or lower too much equipment, tools or materials at a time
- When climbing ladders or scaffoldings, keep three points of contact at all times.
- Stay away from edges of buildings and roofs unless you are secured to a safety line. (Any 3 × 1) (3)

[20]

QUESTION 2

2.1 2.1.1 True

2.1.2 False

2.1.3 True

2.1.4 True

2.1.5 False

2.1.6 True

(6 × 1) (6)

2.2 Consciousness
Airways
Breathing
Circulation (4 × ½) (2)

2.3 Resuscitation is the revival of a person from unconsciousness or apparent death. (1)

2.4 2.4.1
- Dry
- Wet
- Chemical
- Electrical
- Cold
- Radiation (Any 3 relevant answers) (3)

2.4.2
- Dry : fire OR hot material
- Wet : steam OR hot oil/water
- Chemical : acids OR alkaline
- Electrical : lightning OR electricity
- Cold : compressed gas
- Radiation : x-rays OR sunburn (Any 3 relevant answers) (3)

[15]

QUESTION 3

3.1 3.1.1 B

3.1.2 A

3.1.3 E

3.1.4 C

3.1.5 D

(5 × 1) (5)

3.2
- Grip
- Cut
- Bend
- Twist (4)

3.3
- Taper tap
- Intermediate or second tap
- Plug
- Bottoming tap (Any 3 relevant answers) (3)

3.4
- The bending spring must not be stretched.
- The bending spring must be clean.
- The bending spring must not be rusty. (3)

[15]

QUESTION 4

4.1 4.1.1. Portable hand-held electric drilling machine (1)

4.1.2
A Chuck
B Speed-control switch
C Trigger
D Cord
E Trigger lock (5 × 1) (5)

4.1.3 Masonry bit (2)

4.2
- The angle grinder's switch must operate correctly.
- The electrical cord and plug must not be damaged.
- The wheel guard must be secure and in place.
- The handle, disc flange and clamp nut must be tight and not cracked.
- The shaft must turn easily and not be noisy or loose. (Any 4 × 1) (4)

4.3 4.3.1
- Make sure that the sandpaper fits over the base plate correctly and is kept securely in place by proper clamping levers.
- The orbital sander must be clean. (2)

4.3.2 Store the orbital sander in a strong plastic casing or on a specially designed bracket mounted on a tool board in the workshop. (1)

[15]

QUESTION 5

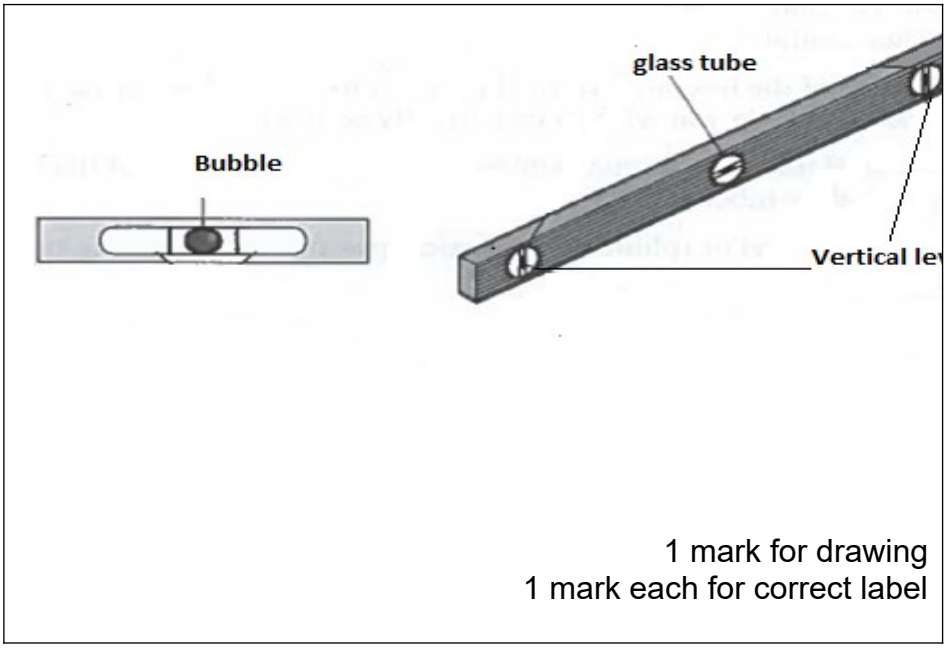
5.1
- Steel ruler
- Measuring tape
- Engineering square
- Spirit level (4)

5.2 5.2.1
- Calibration.
- It must be dust free.
- The jaws must move freely in a firm way.
- The main scale and vernier calliper sliding scale must be legible.
- The locking screw and thumb wheel must be operational.

(Any 5 relevant answer) (5)

5.2.2
- Store the vernier calliper in the enclosure provided by the manufacturer.
- Treat the vernier calliper extremely well as any damage to it will affect the accuracy of measurement. (Any 2 relevant answer) (2)

5.3



1 mark for drawing
1 mark each for correct label

(4)

[15]

QUESTION 6

6.1 The Labour Relations Act deals with the rights of workers✓and the company to ensure a good relationship and a profitable company.✓ (2)

6.2 The purpose of a symbolic safety sign is to achieve instant recognition✓ and understanding by all language and ethnic groups✓ regardless of their degree of literacy.✓ (3)

6.3 6.3.1 Overhead cranes are used to lift and move heavy loads. (1)

6.3.2
- Anchor/bow shackle
- Chain/Dee or D- shackle
- Snap shackle (2 × 1) (2)

6.4 You can waste material and time✓ or not have the correct material and equipment.✓ (2)
[10]

QUESTION 7

7.1 7.1.1 (a) properly soldered

7.1.2 (b) electrical

7.1.3 (c) mechanical

7.1.4 (d) joined
(4 × 1) (4)

7.2 De-soldering is the process of removing the solder from a joint✓ when replacing a faulty component or fixing a dry joint.✓ (2)

7.3 7.3.1 Tinning creates a layer of solder on the solder iron tip, to prevent oxidation (rust). (1)

7.3.2
- All parts must be clean and free from dirt and grease
- Apply a few millimeters of solder to the tip
- Then wipe the bit off on a damp sponge, preparing it for use
(3 ×1) (3)
[10]

TOTAL: 100

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